Week 1

14 February 2026 15:42

Suppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming experience, for that the delay between the action that happens in his phone and the sound heard by him for that action should be as low as possible so that he can play well. Following this scenario, which of the following option(s) is/are correct?

OPTIONS :

- He should buy the earphones which have high latency as much possible.
- It all depends on Internet Speed, so we can't determine anything.
- He should buy the earphones which have low latency as much possible.
- He should buy the earphones which have as low Bluetooth version as possible.

Consider the below server response

HTTP/2.0 500 Internal Server Error

Which of the following is/are **True**?

OPTIONS :

- HTTP/2.0 is the protocol and version respectively
- 500 is the port number
- "Internal Server Error" is the status message
- 500 is the status code
- HTTP/2.0 is the server failure

A client C and the server S located, 12000 km apart are connected with the help of a cable. A request was sent by the client C to server S, but while sending back the response to the client, the cable breaks and the response is now sent back via air. This change in medium caused an additional delay of 20 msec. As compared to the Round Trip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would ____________.

[The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

OPTIONS :

- Increase
- Decrease
- Remain Same
- Insufficient Information

A certain network connection with a bandwidth of 10 Mbps allows making 1000 requests to a server per second. If each request is modified to have an additional data of 8 Kb, what should be the additional bandwidth required to maintain the rate of 2000 requests per second?

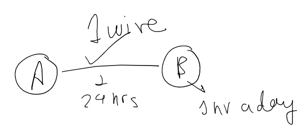
OPTIONS :

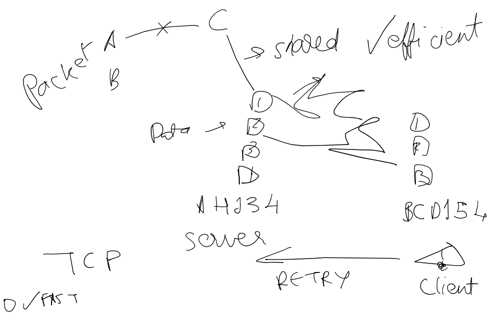
- 18 Mbps
- 16 Mbps
- 26 Mbps
- 36 Mbps

0-255. 0-255. 0-255. 0-255

IPv4 127. 0. 0. 1

IPv6





TCP

10 ✓ FAST

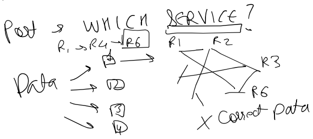
500 ✗ server crashing

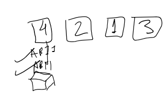
200 ✓

$\boxed{300}$ ✓ server 300 requests / second

IP address → WHICH MACHINE?

Port → WHICH SERVICE?





IP → which machine?

Port → what service on that machine?

TCP → checksum → Data is CORRECT

IP → where packets should go

→ Router 1 → Router 3 → Destination IP address

HTTP → Status code + Body content + Headers

Request is FAIL/SUCCESS?

1. file type?
2. cookie?
3. Language?
4. encoding?

status code → 200 OK

① GET simple send data

② POST receive large data

③ PUT update

④ DELETE

1xx → Information

2xx → SUCCESS

4xx → Client Error (wrong input)

5xx → Backend Error Server crash

3xx → Redirection Moved